The First q-Shifted Pair Theorem: $\det M_R^{(3)} = (\det M_R^{(2)})^2$ in the Hecke Staircase

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Abstract

Let $\Pi_q = B_1 B_2 \cdots B_{n-1}$ be the staircase element in the Iwahori–Hecke algebra $H_q(S_n)$, where $B_j = (1+T_j)(1+T_{j-1})\cdots(1+T_1)$; set $\Pi_q^{(k)} = B_1 \cdots B_k$ for $k \le n-1$, and let $M_R^{(k)}[u,v]$ be the coefficient of T_u in $T_v \cdot \Pi_q^{(k)}$, restricted to the descent-paired set $D_n \subset S_n$. We prove that for every $n \ge 4$,

$$\det M_R^{(3)}[D_n, D_n] \;=\; \big(\det M_R^{(2)}[D_n, D_n]\big)^2$$

as polynomials in $\mathbb{Z}[q]$. As a corollary, the paired-ratio $R_3(n,q)/R_2(n,q) = q^{|D_n|}$ where $|D_n| = n!/2^{\lfloor n/2 \rfloor}$ and $R_k := \det M_R^{(k)} / \det M_R^{(k-1)}$. The proof is by a *locality-reduction* argument: every τ_3-block of D_n is canonically isomorphic to D_4 under an order-preserving identification, reducing the general identity to a direct 6×6 computation at $n = 4$.

1 Setup

Definition 1 (Hecke conventions)**.** Let $H_q(S_n)$ be the Iwahori–Hecke algebra with generators $T_1, \ldots, T_{n-1}$ and relations $T_i^2 = (q-1)T_i + q$ and the usual braid relations. Basis: $\{T_w : w \in S_n\}$. Right multiplication by T_i:

$$T_w \cdot T_i = \begin{cases} T_{ws_i}, & w(i) < w(i+1), \\ q\,T_{ws_i} + (q-1)T_w, & w(i) > w(i+1), \end{cases}$$

where ws_i swaps positions i and $i+1$ in the one-line notation of w.

Definition 2 (Staircase and matrices)**.** $B_j := (1+T_j)(1+T_{j-1})\cdots(1+T_1)$ and $\Pi_q^{(k)} := B_1 B_2 \cdots B_k$. The full staircase is $\Pi_q := \Pi_q^{(n-1)}$.

Definition 3 (Descent-paired set)**.** $D_n = \{w \in S_n : w(2i-1) > w(2i) \text{ for all } i = 1, \ldots, \lfloor n/2 \rfloor\}$. One has $|D_n| = n!/2^{\lfloor n/2 \rfloor}$.

Definition 4 (Staircase matrix)**.** For $u, v \in D_n$ and $k \in \{1, \ldots, n-1\}$, set

$$M^{(k)}[u,v] \;:=\; [T_u]\big(T_v \cdot \Pi_q^{(k)}\big),$$

the coefficient of T_u in the expansion of $T_v \cdot \Pi_q^{(k)}$. This defines a $|D_n| \times |D_n|$ matrix over $\mathbb{Z}[q]$.

We write $\tau_k(w) := (w(k+2), \ldots, w(n))$ for the length-$(n-k-1)$ suffix. The τ_3-blocks of D_n are the preimages of τ_3 on D_n; we show below that $\Pi_q^{(k)}$ preserves this partition for $k \le 3$.

2 Main Theorem

Theorem 5 (First q-Shifted Pair)**.** *For every $n \geq 4$,*

$$\det M^{(3)}[D_n, D_n] \;=\; \big(\det M^{(2)}[D_n, D_n]\big)^2 \quad \text{in } \mathbb{Z}[q].$$

Corollary 6. *With $R_k(n,q) := \det M^{(k)}[D_n, D_n] / \det M^{(k-1)}[D_n, D_n]$,*

$$R_3(n,q) \;=\; q^{|D_n|} \cdot R_2(n,q), \qquad |D_n| = \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Proof of Corollary 6 given Theorem 5. $R_3/R_2 = \det M^{(3)} \det M^{(1)} / (\det M^{(2)})^2 = \det M^{(1)}$ by Theorem 5. For $w \in D_n$, $w(1) > w(2)$, so $T_w \cdot (1 + T_1) = T_w + qT_{ws_1} + (q-1)T_w = qT_w + qT_{ws_1}$; the T_w-coefficient is q. Since $ws_1 \notin D_n$ (as $ws_1(1) = w(2) < w(1) = ws_1(2)$), no off-diagonal contributions land back on D_n; therefore $M^{(1)}[D_n, D_n]$ is diagonal with entries all equal to q, and $\det M^{(1)} = q^{|D_n|}$. □

3 Proof of Theorem 5

The proof consists of three locality lemmas plus a base case at $n = 4$.

3.1 Locality

Lemma 7 (Locality of $\Pi^{(k)}$ for $k \leq 3$)**.** *For $k \leq 3$, $\Pi_q^{(k)}$ lies in the subalgebra $\langle T_1, T_2, T_3 \rangle \subset H_q(S_n)$. In particular, for any $w \in S_n$, the support of $T_w \cdot \Pi_q^{(k)}$ in the T-basis consists of $T_{w'}$ with $w'(i) = w(i)$ for all $i \geq 5$.*

Proof. By definition $\Pi_q^{(k)} = B_1 \cdots B_k$ where B_j is a product of factors $(1+T_i)$ with $i \leq j \leq k \leq 3$, so all factors use only T_1, T_2, T_3. The right-action of T_i on T_w produces terms supported on $\{T_w, T_{ws_i}\}$; when $i \leq 3$, ws_i swaps positions $i, i+1$ in one-line notation, leaving positions $5, \ldots, n$ unchanged. Iterating through the factors of $\Pi_q^{(k)}$ preserves this property. □

Lemma 8 (τ_3-block structure)**.** *For each descent-paired suffix $\sigma' = (a_5, \ldots, a_n)$ (i.e., satisfying $a_{2i-1} > a_{2i}$ for all $i \geq 3$), the τ_3-block $V_{\sigma'} := \{w \in D_n : \tau_3(w) = \sigma'\}$ has exactly $6 = |D_4|$ elements, indexed by the $|D_4|$ descent-paired arrangements of the prefix-value set $P_{\sigma'} := \{1, \ldots, n\} \setminus \{a_5, \ldots, a_n\}$.*

Proof. The defining descent-pair constraints of D_n split into prefix constraints ($w(1) > w(2)$ and $w(3) > w(4)$) and suffix constraints ($a_{2i-1} > a_{2i}$ for $i \geq 3$). For fixed valid σ' the suffix constraints hold by hypothesis; the prefix is an arbitrary arrangement of $P_{\sigma'}$ subject to the two prefix constraints, yielding $4!/2^2 = 6$ choices. □

Lemma 9 (Block isomorphism)**.** *Let σ' be a valid suffix and $P_{\sigma'} = \{p_1 < p_2 < p_3 < p_4\}$ its prefix-value set. Let $\phi_{\sigma'} : P_{\sigma'} \to \{1, 2, 3, 4\}$ be the unique order-preserving bijection. Define the linear map*

$$\Phi_{\sigma'} : \operatorname{span}(T_w : w \in V_{\sigma'}) \longrightarrow \operatorname{span}(T_u : u \in D_4), \qquad T_w \mapsto T_{\phi_{\sigma'} \circ (w|_{\{1,2,3,4\}})}.$$

Then $\Phi_{\sigma'}$ is a $\mathbb{Z}[q]$-module isomorphism that intertwines right-multiplication by $\Pi_q^{(k)}$ for every $k \leq 3$.

Proof. Bijection. By Lemma 8, $V_{\sigma'}$ is in bijection with the descent-paired arrangements of $P_{\sigma'}$; $\phi_{\sigma'}$ then gives a bijection with D_4 (descent-paired arrangements of $\{1,2,3,4\}$), since $\phi_{\sigma'}$ is order-preserving.

Intertwining. It suffices to check intertwining with right-multiplication by each T_i, $i \in \{1,2,3\}$. For $w \in V_{\sigma'}$: if $w(i) < w(i+1)$, then $T_w T_i = T_{ws_i}$; since $\phi_{\sigma'}$ preserves order, $\phi_{\sigma'}(w(i)) < \phi_{\sigma'}(w(i+1))$, so on the D_4 side $\Phi_{\sigma'}(T_w) \cdot T_i$ uses the same rule. Since $i \leq 3$, ws_i only changes positions $i, i+1$ (both in $\{1,2,3,4\}$), so $\Phi_{\sigma'}(T_{ws_i}) = \Phi_{\sigma'}(T_w) \cdot T_i$ with coefficient 1. The $w(i) > w(i+1)$ case is analogous, with the same coefficients q and $q-1$. The result extends to products by linearity and associativity. □

3.2 Reduction to $n = 4$

Lemma 10 (Global factorization)**.** *For $k \in \{1,2,3\}$,*

$$\det M^{(k)}[D_n, D_n] \;=\; \big(\det M^{(k)}[D_4, D_4]\big)^{N(n)},$$

where $N(n) = |D_n|/6 = \binom{n}{4} \cdot |D_{n-4}|$ is the number of valid τ_3-blocks.

Proof. Since $\Pi_q^{(k)}$ fixes positions $5, \ldots, n$ (Lemma 7), it preserves the τ_3-block partition of D_n; hence $M^{(k)}[D_n, D_n]$ is block-diagonal with respect to $\{V_{\sigma'}\}_{\sigma'}$, giving

$$\det M^{(k)}[D_n, D_n] \;=\; \prod_{\sigma' \text{ valid}} \det M^{(k)}\big|_{V_{\sigma'}}.$$

By Lemma 9, $\det M^{(k)}|_{V_{\sigma'}} = \det M^{(k)}[D_4, D_4]$ for every valid σ'; the count of valid suffixes equals $N(n)$ (a descent-paired sequence of length $n-4$ using any $(n-4)$-subset of $\{1, \ldots, n\}$). □

3.3 Base case: $n = 4$

Lemma 11. $\det M^{(3)}[D_4, D_4] = \big(\det M^{(2)}[D_4, D_4]\big)^2 = q^{18}(q+1)^6(2q+1)^2$.

Proof. Direct computation (exact SymPy over $\mathbb{Q}(q)$; see `/home/clio/projects/scratch/xmx_final.py`). The τ_2-block decomposition of D_4 has sizes $(3,2,1)$ (indexed by $w(4) = 1,2,3$); by Rule B, $M^{(2)}[D_4, D_4]$ is block-diagonal with determinants

$$\begin{aligned}
\det M^{(2)}|_{\text{size } 3} &= q^5(q+1)^2, \\
\det M^{(2)}|_{\text{size } 2} &= q^3(2q+1), \\
\det M^{(2)}|_{\text{size } 1} &= q(q+1).
\end{aligned}$$

Their product is $\det M^{(2)}[D_4, D_4] = q^9(q+1)^3(2q+1)$.

The single τ_3-block of D_4 is all of D_4 (size 6); direct computation of the 6×6 determinant yields

$$\det M^{(3)}[D_4, D_4] = q^{18}(q+1)^6(2q+1)^2 = \big(q^9(q+1)^3(2q+1)\big)^2.$$

□

3.4 Assembly

Proof of Theorem 5. By Lemma 10 applied to $k = 2$ and $k = 3$:

$$\det M^{(3)}[D_n, D_n] = \big(\det M^{(3)}[D_4, D_4]\big)^{N(n)},$$
$$\det M^{(2)}[D_n, D_n] = \big(\det M^{(2)}[D_4, D_4]\big)^{N(n)}.$$

By Lemma 11, $\det M^{(3)}[D_4, D_4] = (\det M^{(2)}[D_4, D_4])^2$. Raising both sides to the $N(n)$-th power gives

$$\det M^{(3)}[D_n, D_n] = \big(\det M^{(2)}[D_n, D_n]\big)^2.$$

□

4 Computational verification

All steps of the proof were verified by exact symbolic computation over $\mathbb{Z}[q]$ in SymPy:

- $n = 4$ (Lemma 11): direct 6×6 determinant. See `xmx_final.py`; output matches $\det M^{(3)} = q^{18}(q+1)^6(2q+1)^2$.
- $n = 5$ (locality): all 5 valid τ_3-blocks of D_5 are IDENTICAL as 6×6 matrices after the order-preserving relabeling of $\phi_{\sigma'}$. Each has determinant $q^{18}(q+1)^6(2q+1)^2$, and the collective $\det M^{(2)}$ equals $(q^9(q+1)^3(2q+1))^5$.
- $n = 5, 6, 7$ (main identity): $\det M^{(3)}[D_n] - (\det M^{(2)}[D_n])^2 = 0$ as polynomials (verified by `sp.expand` and direct comparison). See `prove_locality_check.py`.

5 Remarks

Remark 12 (Correction of the original conjecture)**.** The originally conjectured form $R_3(n, q) = q^{6\binom{n}{4}} R_2(n, q)$ is numerically consistent with Corollary 6 for $n \in \{4, 5, 6\}$ because $|D_n| = 6\binom{n}{4}$ for these values, but the two expressions *diverge* at $n = 7$ and beyond:

$$|D_7| = 630, \qquad 6\binom{7}{4} = 210.$$

Explicit computation confirms $R_3(7, q)/R_2(7, q) = q^{630}$, not q^{210}. The combinatorial cleanliness of the statement $\det M^{(3)} = (\det M^{(2)})^2$ (and its corollary with exponent $|D_n|$) appears to be the natural formulation.

Remark 13 (Why the identity $\det M^{(3)}[D_4). = (\det M^{(2)}[D_4])^2$ at $n = 4$?] Theorem 5 reduces to a direct computational check at $n = 4$. The numerological coincidence that $\Pi_q^{(3)}$ for $n = 4$ equals the full staircase Π_q of $H_q(S_4)$, while $\Pi_q^{(2)}$ is the one-step-short truncation, suggests a deeper structural reason — perhaps related to the eigenvalue palindromy of the staircase operator, or to a duality between $\Pi^{(k)}$ and $\Pi^{(n-1-k)}$. We leave this as an open question (Section 6).

Remark 14 (Higher k)**.** The locality argument extends: for any fixed k, $\Pi_q^{(k)}$ lies in $\langle T_1, \ldots, T_k \rangle \cong H_q(S_{k+1})$, and the τ_k-block structure of D_n reduces to D_{k+1} under order-preserving relabeling. Hence all identities among $\{\det M^{(j)}[D_n]\}_{j \le k}$ reduce to identities at $n = k + 1$. This suggests a systematic program for higher pairs R_{2j}/R_{2j+1}.

6 Open questions

1. **Structural reason for the $n = 4$ squaring.** Is there a proof of $\det M^{(3)}[D_4] = (\det M^{(2)}[D_4])^2$ that avoids direct computation? Eigenvalue palindromy and the bicoboundary (Apr 17) result may be relevant.

2. **Higher pairs.** What is the analogous identity for $\det M^{(5)}$ vs. $\det M^{(4)}$? By the locality program it reduces to $n = 6$, but $|D_6| = 90$; exact computation is feasible but nontrivial.

3. **Non-paired ratios.** What is R_2/R_3 as a polynomial in q for generic n? Our corollary gives $R_3 = q^{|D_n|}R_2$, but the individual R_k's have rich structure involving $(q+1)$ and $(2q+1)$ factors with multiplicities $3|D_n|/6$ and $|D_n|/6$ respectively.